SALEM NASSAR

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Profile

Software engineer with 2+ years at Verizon building Python automation and OpenShift infrastructure that runs in production. Outside of work, I build autonomous systems in PyTorch and C++, including a self-driving RC car that runs C++ inference on a Raspberry Pi. Currently pursuing an M.S. in Computer Science with a focus in Machine Learning at Georgia Tech.

Experience

Software Engineer June 2024 – Present
Verizon, Southlake, TX

- Built 20+ Python automations that generate and send router configs for sites and access leaves nationwide across 8+ vendor platforms (Cisco, Nokia, Ericsson, Samsung, Ribbon)
- Reduced deployment time by 90% with GitLab CI/CD multi-stage Docker builds and OpenTofu-automated multi-cluster OpenShift deployments
- Solely deployed Nautobot and HashiCorp Vault to 5 sites with OpenTofu, automating multi-site OpenShift rollouts for both

Projects

Autonomous RC Car: Sim-Trained RL Driving Policy

- Built a GPU-parallel **MuJoCo Warp** sim of my RC car (1,024 cars on 1,024 random tracks, ~17.5K camera-rendered steps/s) and trained a camera-only **PilotNet** actor with a state critic using **PPO**
- Added **domain randomization** of latency (40–75 ms), servo trim, speed, steering, floor texture, lane-line color, camera pose, exposure, and sensor noise
- A smooth-steering reward cut steering change per step by ~50% in a 600-iteration run (~19 min); on **256 held-out tracks** the policy finished **100%** without crashing (**91%** with randomized car and visuals)
- Trained the on-car PilotNet via **behavioral cloning** on **10K+ frames** (drives the track **100% autonomously**); multithreaded **C++** ONNX Runtime inference in **3.6 ms** on Raspberry Pi 5
- In progress: deploying the sim-trained policy on the real car

Parallel Cartpole (GPU Reinforcement Learning)

- Built a GPU-parallel RL environment with **MuJoCo Warp** and custom **NVIDIA Warp** kernels, capturing each step as a **CUDA graph** to reach **18.8M env steps/s** across 16K simulations
- Implemented **PPO** from scratch in PyTorch with an asymmetric actor-critic, training a CNN on rendered camera frames alone to balance the pole in **34 s** on one GPU

Technical Skills

- **Machine Learning and Computer Vision:** PyTorch, ONNX, NumPy, OpenCV, CNNs, Behavioral Cloning, Reinforcement Learning (PPO), Scikit-learn
- **Simulation and GPU:** MuJoCo, MuJoCo Warp, NVIDIA Warp, CUDA Graphs, Mixed Precision (bf16)
- **Programming Languages:** Python, C++, C, Bash
- **DevOps and Automation Tools:** Kubernetes, OpenShift, Docker, Helm, GitLab CI/CD, OpenTofu/Terraform, Linux, Git, Vault
- **Databases:** PostgreSQL, SQL, Redis, MongoDB
- **AI-Assisted Development:** Claude Code, Codex, Grok Build

Education

Master of Science in Computer Science, Specialization in Machine Learning Aug 2025 – May 2028
Georgia Institute of Technology (Online)

- GPA: 4.0
- Completed: Artificial Intelligence, Computer Vision, Machine Learning, AI Techniques for Robotics (SLAM, localization, motion planning), Reinforcement Learning
- Current: Deep Learning

Bachelor of Science in Computer Science May 2024
The University of Texas at Dallas